File

Name: P4-fit-A1
Description: KISSsoft example
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1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

Diameter shaft (mm)	[DiA]/[DiI]	50.00	/ 0.00
Diameter hub (mm)	[DaA]/[DaI]	100.00	/50.00
Length of Interference fit (mm)	[l]	40.00	
Diameter of joint (mm)	[DF]	50.00	
with special allowances:			
Upper allowance Shaft (µm)	[AoI]	60.6	
Lower allowance Shaft (µm)	[AuI]	60.6	
Tolerance measure Shaft (µm)	[TI]	0.0	
Upper allowance Hub (µm)	[AoA]	0.0	
Lower allowance Hub (µm)	[AuA]	0.0	
Tolerance measure Hub (µm)	[TA]	0.0	
Largest interference (µm)	[Po]	60.6	
Smallest interference (µm)	[Pu]	60.6	
Nominal torque (Nm)	[T]	0.00	
Application factor	[KA]	1.00	
Service torque (Nm)	[Tb]	0.00	
Axial force (N)	[FA]	0.00	
Bending moment (Nm)	[Mb]	0.00	
Radial force (N)	[Fr]	0.00	
Circumferential force (N)	[Fu]	0.00	
Speed (1/min)	[n]	0.00	
Interference (µm)	[P]	60.6	(60.6 .. 60.6)
Embedding (µm)	[s]	3.84	
Effective interference at 20°C(68°F) (µm)	[Pw]	56.77	(56.8 .. 56.8)
Effective interference at working temp.. (µm)	[PwTh]	56.77	(56.8 .. 56.8)
Service temperature shaft (°C)	[ThB]	20.0	
Service temperature hub (°C)	[ThB]	20.0	
Pressure stress by			
- Interference (after mounting) (N/mm²)	[pM]	91.55	(91.55 .. 91.55)
- Interference (working) (N/mm²)	[p]	91.55	(91.55 .. 91.55)
- Bending moment (N/mm²)	[pb]	0.00	
- Radial force (N/mm²)	[pr]	0.00	
Coefficient of friction, longitudinal	[µa]	0.120	
Coefficient of friction, circumferential	[µu]	0.120	
Safety against sliding	[Sr]	> 1000	
Required safety against sliding	[SSr]	1.50	

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

1.1 Shaft

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[El]	215000.00			
Poisson's ratio (-)	[vl]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (component's raw diameter) (N/mm²)	[Rm]	650.00			
Yield point (component's raw diameter) (N/mm²)	[Rp]	200.00			
Component raw diameter (mm)	[deff]	55			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DiA]	50.00			
Diameter increase during operation (µm)	[ΔD]	-14.90	(-14.90 ..	-14.90	)
Diameter increase during operation (o/oo)	[ΔD]	-0.30	(-0.30 ..	-0.30	)
Equivalent stress outer diameter (N/mm²)	[σVa]	91.55	(91.55 ..	91.55	)
- Radial stress (N/mm²)	[σRa]	-91.55	(-91.55 ..	-91.55	)
- Tangential stress (N/mm²)	[σTa]	-91.55	(-91.55 ..	-91.55	)
- With outside load (N/mm²)	[σVaMF]	91.55	(91.55 ..	91.55	)
Inner diameter (mm)	[Dil]	0.00			
Equivalent stress inner diameter (N/mm²)	[σVi]	91.55	(91.55 ..	91.55	)
- Radial stress (N/mm²)	[σRi]	-91.55	(-91.55 ..	-91.55	)
- Tangential stress (N/mm²)	[σTi]	-91.55	(-91.55 ..	-91.55	)
Safety against fracture	[SiRm]	7.10	(7.10 ..	7.10	)
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Rp]	2.18	(2.18 ..	2.18	)
Required safety against yield point	[SSi.Rp]	1.10			

1.2 Hub

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[EA]	215000.00			
Poisson's ratio (-)	[vA]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	700.00			
Yield point (wall thickness) (N/mm²)	[Rp]	300.00			
The hub material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	26			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DaA]	100.00			
Diameter increase during operation (µm)	[ΔD]	28.39	(28.39 ..	28.39	)
Diameter increase during operation (o/oo)	[ΔD]	0.28	(0.28 ..	0.28	)
Equivalent stress outer diameter (N/mm²)	[σVa]	61.03	(61.03 ..	61.03	)
- Radial stress (N/mm²)	[σRa]	0.00	(0.00 ..	0.00	)
- Tangential stress (N/mm²)	[σTa]	61.03	(61.03 ..	61.03	)
Inner diameter (mm)	[DaI]	50.00			
Diameter increase during operation (µm)	[ΔD]	41.87	(41.87 ..	41.87	)
Diameter increase during operation (o/oo)	[ΔD]	0.84	(0.84 ..	0.84	)
Equivalent stress inner diameter (N/mm²)	[σVi]	213.61	(213.61 ..	213.61	)
- Radial stress (N/mm²)	[σRi]	-91.55	(-91.55 ..	-91.55	)
- Tangential stress (N/mm²)	[σTi]	152.58	(152.58 ..	152.58	)
- Radial stress with outside load (N/mm²)	[σRai]	-91.55	(-91.55 ..	-91.55	)
- Tangential stress with outside load (N/mm²)	[σTai]	152.58	(152.58 ..	152.58	)
- With outside load (N/mm²)	[σViMF]	213.61	(213.61 ..	213.61	)

Safety against fracture	[Si.Rm]	3.28	(	3.28	..	3.28)
Required safety against fracture	[SSi.Rm]	1.50				
Safety against yield point	[Si.Re]	1.40	(	1.40	..	1.40)
Required safety against yield point	[SSi.Re]	1.10				

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

2 Service / Mounting / Dismounting

Transverse-interference-fit:						
Mounting clearance (mm)	[PsTh]	0.050				
Temperature difference for mounting:						
Shaft temperature: (°C) Hub temperature: [ThA] (°C)						
Mount with shaft temperature 20 °C:						
Assembly temperature shaft (°C)	[θI]	20				
Assembly temperature hub (°C)	[θA]	212.372				
Mount with shaft temperature -150 °C:						
Assembly temperature shaft (°C)	[θI]	-150				
Assembly temperature hub (°C)	[θA]	86.720				
calculated using the coefficient of thermal expansion						
Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K)	[αw]	8.50				
Longitudinal pressure fit:						
Assembly temperature shaft (°C)	[θM]	20.00				
Assembly temperature hub (°C)	[θM]	20.00				
Coefficient of friction, longitudinal	[μe=μa*	1.300]	0.16			
Press on (force) (kN)	[Fpress]	89.73	(	89.73	..	89.73)
Coefficient of friction, longitudinal	[μl=μa*1.600]	0.19				
Press out (force) (kN)	[Fpress]	110.44	(	110.44	..	110.44)
Coefficient of friction	[μ]	0.19				
Max. torque to avoid Micro sliding (Nm)	[Tlimit]	528.34	(	528.34	..	528.34)

Notice concerning the display: Number-1 (Number-2.. Number-3):
- Number-1: Value calculated with the mean allowance
- Number-2: Value of the smallest possible allowance
- Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

End of report (lines: 154)